

# LEA C. MICHEL

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## ACADEMIC POSITIONS

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### Postdoc in Predictive Clinical Neuroscience Lab

Donders Institute for Brain, Cognition, and Behaviour

*May 2026 - June 2027*

*Nijmegen, NL*

Supervisors: Professor Andre Marquand

Project: Precognition: Learning Latent Cognitive Profiles to Predict Psychosis

### Ph.D. in the Lifespan Cognitive Dynamics Lab

Donders Institute for Brain, Cognition, and Behaviour

*Sept 2021 - March 2026*

*Nijmegen, NL*

Supervisors: Professor Rogier Kievit and Professor Carolina de Weerth

Thesis: Understanding the interplay between grey and white matter structure in supporting cognitive development in childhood and adolescence

### Visiting researcher

Lifespan Modelling and Psychometrics Lab, King's College London

*Jan-April 2025*

*London, UK*

Supervisor: Dr Delia Fuhrmann

Subject: The neural scars of adversity: how different kinds of adversity shape the development of grey and white matter in the brain

### Visiting researcher

Social, Cognitive, Affective and Neuroscience, University of Birmingham

*March 2023*

*Birmingham, UK*

Supervisor: Professor Stephane De Brito

Subject: Finishing the VBM project on youths with conduct disorder the effect of gender and callous-unemotional traits

### Internship at Addiction and Mental Health Group

Department of Psychology, University of Bath

*Feb 2021 - Jul 2021*

*Bath, UK*

Supervisor: Dr Esther Walton

MSc2 Thesis: The role of epigenetic age in the relationship between stressful events and brain structure in adolescent males of the ALSPAC cohort

### Internship at Developmental Biopsychiatry Research Program

McLean Hospital

*Mar 2020 - Jul 2020*

*Boston, USA*

Supervisors: Dr Kyoko Ohashi and Dr Martin Teicher

Subject: Effect of childhood maltreatment on functional connectivity in resting-state and MIST task

### Internship at the SCAN lab

Social, Cognitive, Affective and Neuroscience, University of Birmingham

*Aug 2019 - Jan 2020*

*Birmingham, UK*

Supervisor: Dr Stephane De Brito

Subject: Voxel-Based Morphometry study on youths with conduct disorder the effect of sex and callous-unemotional traits

**Internship at the Memory, Brain and Cognition laboratory** *Nov 2018 - Jun 2019*  
Memory, Brain and Cognition laboratory (LMC<sup>2</sup>), Psychology Department, Université de Paris Boulogne-Billancourt, France

Supervisors: Dr Pascale Piolino and Dr Valentina La Corte  
MSc1 Thesis: Study of the neural basis of episodic memory

**Internship at the Babylab** *Jun - Jul 2018*  
Laboratoire de Sciences Cognitives et Psycholinguistiques, ENS Ulm Paris, France

Supervisor: Dr Mireille Babineau  
Subject: Synergies in the acquisition of syntactic categories

## EDUCATION

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**Ph.D. in Developmental Neuroscience** *Sept 2021 - March 2026*  
Donders Institute for Brain, Cognition, and Behaviour, RadboudUMC *Nijmegen, NL*

**MSc in Cognitive Sciences - major in Neuroscience** *Sept 2018 - June 2021*  
Ecole Normale Supérieure, Université de Paris, EHESS *Paris, FR*  
Gap year from September 2019 to August 2020 to do internships

**BSc in Health Sciences - major in Neuroscience** *Sept 2017 - June 2018*  
Université de Paris *Paris, FR*

**Higher School Preparatory Class in Biology** *Sept 2014 - June 2017*  
Lycée Faidherbe *Lille, FR*  
Two-year undergraduate programme (120 ECTS) preparing for the Grandes Ecoles competitive exams. Competitive admission (<10% of students admitted) and one of the highest workload in Europe (~45 contact hours and up to 10 hours of oral and written exams per week )

## PUBLICATIONS

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Carrick, C., Rakesh, D., **Michel, L.C.**, L., Bates, K., Fuhrmann, D. (2026). Individual differences in adolescent cortical development are associated with neighbourhood characteristics: Longitudinal findings from the ABCD study. *Cerebral Cortex*, 36(4), bhag034.

Kalc, P., Rae, C., Tansey, R., Wang, M., Nieto, N., Komeyer, V., ... ,**Michel, L.C.**, Souter, N. Hinault, T. (2026). The Organization for Human Brain Mapping online Satellite meeting 2025: Enhancing accessibility and sustainability. *Aperture Neuro*, 6.

Sadeghi, N., Van Der Velpen, I., Baker, B., Batta, I., Genon, S., McCormick, E. M., **Michel, L.C.**, ... , White, T. (2025). The interplay between brain and behavior during development: A multisite effort to generate and share simulated datasets. *Scientific Data*, 12(1), 473.

Coolen, I. E., van Langen, J., Hofman, S., van Aagten, F. E., Schaaf, J. V., **Michel, L.C.**, Aristodemou, M., Judd, N., van Hout, A. T. B., Meeussen, E., Kievit, R. A. (2024). Protocol and preregistration for the CODEC project: measuring, modelling and mechanistically understanding the nature of cognitive variability in early childhood. *BMC psychology*, 12(1), 407.

**Michel, L.C.**, McCormick, E. M., Kievit, R. A. (2024). Grey and white matter metrics demonstrate distinct and complementary prediction of differences in cognitive performance in children: Findings from ABCD (N= 11 876). *Journal of Neuroscience*.

Nikolaidis, A., Mancini, M., Auer, T., Bottenhorn, K. L., ..., **Michel, L.C.**, ... Zwiers, M. P. (2023). Proceedings of the OHBM Brainhack 2021. *Aperture Neuro*, 3.

### **In Preparation (thesis chapters)**

**Michel, L.C.**, Sørensen, Ø., McCormick, E. M., Parsons, S., Madsen, K. S., Baare, W., Kievit, R. A. Grey and white matter development is surprisingly independent: a longitudinal study in a 12-wave sample.

**Michel, L.C.**, Judd, N., de Kock, C. P.J., Goriounova, N., Kievit, R. A. Bridging the gap: Integrating histological and MRI data to better understand brain-behavior relationships.

**Michel, L.C.**, Rakesh, D., Whelan, R., Fuhrmann, D., Kievit, R. A. and the IMAGEN consortium. The neural scars of adversity: how do different kinds of adversity predict the development of grey and white matter structure in adolescence?

## **SCHOLARSHIPS AND AWARDS**

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### **Awards Radboud Young Academy Recognises and Rewards prize** *June 2026*

Nomination for the Radboud Young Academy Recognises and Rewards prize for an employee who contributes to a healthy and effective work environment

### **Donders Cube Award** *May 2026*

The Donders Cube is awarded annually to a person or a team of persons who improve the institute in a general sense.

### **Flux Travel Award** *September 2025*

Grant to support my visit to the Flux conference 500 \$

### **Radboud Young Academy Recognises and Rewards prize** *June 2025*

Nomination for the Radboud Young Academy Recognises and Rewards prize for an employee who contributes to a healthy and effective work environment

### **PhD talk prize - Donders Discussion** *October 2024*

Prize for the best individual talk during the Donders Discussion (Donders conference for PhD)

### **Cognitive Neuroscience Department Team Player Award** *2024*

Award recognizing personal contribution to the department beyond of what can be expected from their position

### **Grants Radboud Internationalisation Travel Grant** *January 2025*

Grants for outgoing PhD to support my research visit in London 700€

### **Society for the Improvement of Psychological Science Grant** *2023*

Grants-In-Aid to Reduce Barriers to Improving Psychological Science 300 \$

### **Erasmus Staff Training Grant** *2023*

2 weeks internship at the CHBH at the University of Birmingham approx 2,000€

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|------------------------------------------------------------------------------------------------|------------------|
| <b>MIBPT – Midlands Integrative Biosciences Training Partnership</b>                           | <b>2021</b>      |
| BBSRC funded Doctoral Training Partnership between 5 universities                              | approx £76,000   |
| 1st candidate accepted in the Psychology department at the University of Birmingham (declined) |                  |
| <b>MIRES – Excellence mobility scholarship for master students</b>                             | <b>2021</b>      |
| Universite de Paris                                                                            | 5,000€           |
| <b>Minister of Higher Education and Research scholarship</b>                                   | <b>2020</b>      |
| Universite de Paris                                                                            | 2,000€           |
| <b>MIRES – Excellence mobility scholarship for master students</b>                             | <b>2019-2020</b> |
| Universite de Paris                                                                            | 5,000€           |

## TEACHING EXPERIENCE

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| <b>Instructor - Workshop on Latent Growth Curve Model</b>                                                                       | <b>Nov 2024</b>     |
| Designing and teaching an afternoon workshop for researchers from a consortium on latent growth curve model                     | <i>Nijmegen, NL</i> |
| Donders Institute                                                                                                               |                     |
| <b>Instructor - Modelling Neurodevelopment using Big Data</b>                                                                   | <b>2023-2024</b>    |
| Designing and teaching a two-days course for BSc students about R, regressions, plots and neurodevelopment                      | <i>Nijmegen, NL</i> |
| Donders Institute                                                                                                               |                     |
| <b>Teaching Assistant - Early Life: The Key to Preventing Lifelong Physical and Mental Disease</b>                              | <b>2021-2023</b>    |
| Coordinating the BSc course and grading the assignments and exams                                                               | <i>Nijmegen, NL</i> |
| Donders Institute                                                                                                               |                     |
| <b>Teaching Assistant - Early onset neuropsychiatric disorders: Brain mechanisms and genetic and environmental risk factors</b> | <b>2021-2023</b>    |
| Coordinating the MSc course                                                                                                     | <i>Nijmegen, NL</i> |
| Donders Institute                                                                                                               |                     |
| <b>Teacher: Introduction to Psychology</b>                                                                                      | <b>Jul, 2021</b>    |
| Designing and teaching a psychology course for 4 weeks for a group of international students from grade 9 to 12                 | <i>Paris, FR</i>    |
| Oxbridge in Paris – Oxbridge Academic programs                                                                                  |                     |

## SUPERVISION

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|-------------------------------------------------------------------------------|-----------------------|
| <b>Supervising MSc first year</b>                                             | <b>Oct - Dec 2024</b> |
| Fernando Fernández de Cuevas - Cognitive Neuroscience MSc                     | Donders Institute, NL |
| Quality control of structural MRI scans in patients                           |                       |
| <b>Supervising MSc final year research project</b>                            | <b>2023 - 2024</b>    |
| Nebbe Al-Moula - Cognitive Neuroscience MSc                                   | Donders Institute, NL |
| "Sex Specific Associations Between Brain Structure and Autism Symptomatology" |                       |

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| <b>Supervising MSc final year research project</b>                                                                        | <b>2023 - 2024</b>    |
| Liza Rozman - Cognitive Neuroscience MSc                                                                                  | Donders Institute, NL |
| "ADHD Symptom Reduction Enhances White Matter Fractional Anisotropy Without Altering Sleep Latency: A Longitudinal Study" |                       |
| <b>Supervising BSc grant proposal research project</b>                                                                    | <b>2023 - 2024</b>    |
| Writing grant proposals with a group of five biomedical BSc students                                                      | Donders Institute, NL |
| "Co-development of grey and white matter from birth to 3 years old in premature infants"                                  |                       |
| <b>Supervising BSc grant proposal research project</b>                                                                    | <b>2023 - 2024</b>    |
| Writing grant proposals with a group of five biomedical BSc students                                                      | Donders Institute, NL |
| "Potential early biomarkers for autism in high-risk infants: a longitudinal study"                                        |                       |

## TALKS

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### Conferences

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| <b>The neural scars of adversity: How do different kinds of adversity predict the development of grey and white matter structure in two longitudinal neuroimaging cohorts?</b> | <b>September, 2025</b> |
| Flash talk at the Flux Congress                                                                                                                                                | Dublin, IR             |
| <b>Grey or white, does it matter?</b>                                                                                                                                          | <b>June, 2025</b>      |
| Individual talk at the Donders Day                                                                                                                                             | Nijmegen, NL           |
| <b>Bridging the gap: Integrating histological and MRI data to better understand brain-behavior relationships</b>                                                               | <b>October, 2024</b>   |
| Individual talk at the Donders Discussion                                                                                                                                      | Nijmegen, NL           |
| <b>Grey and white matter are not two separate tissues</b>                                                                                                                      | <b>October, 2023</b>   |
| Lightning talk at the Donders Discussion                                                                                                                                       | Nijmegen, NL           |
| <b>The longitudinal development of grey and white matter metrics in typical development</b>                                                                                    | <b>March, 2023</b>     |
| International Convention of Psychological Science (ICPS)                                                                                                                       | Brussels, BE           |
| Symposium on "Longitudinal Neuroimaging As a Key Avenue to Understanding Cognitive Development: Recent Findings across Four Cohorts."                                          |                        |
| <b>Does grey and white matter microstructure play complementary roles in supporting cognitive ability ?</b>                                                                    | <b>April, 2022</b>     |
| Talk at the Colloquium of Young Researchers in Cognitive Sciences in France                                                                                                    | Paris, FR              |
| <b>Grey Matter Volume in Youths with Conduct Disorder: Effects of Callous-Unemotional Traits</b>                                                                               | <b>March, 2021</b>     |
| Talk at the Colloquium of Young Researchers in Cognitive Sciences in France                                                                                                    | (virtual)              |

### Invited talks

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|---------------------------------------|------------------|
| <b>Grey or white, does it matter?</b> | <b>Mar, 2025</b> |
| Neuroscience Sussex Seminar           | Brighton, UK     |

**Understanding the interplay between grey and white matter structure in supporting cognitive development in childhood and adolescence** *Mar, 2025*  
Birmingham, UK  
Lab meeting in the SCAN lab, Center for Human Brain Health

**Understanding the interplay between grey and white matter structure in supporting cognitive development in childhood and adolescence** *Feb, 2025*  
London, UK  
Lab meeting in the LIFE-MAP lab, King's College London

**The interplay between grey and white matter during typical development** *Oct, 2023*  
Birmingham, UK  
Visit meeting in the Center for Human Brain Health

## Department

**Green Team: Exploring a more sustainable academia** *April, 2025*  
Donders, NL  
Department meeting

**Sustainability in academia : Summary of the achievement of the Green Team within the department** *March, 2024*  
Donders, NL  
Department meeting

**Understanding the interplay between grey and white matter structure cognitive development in childhood and adolescence** *Oct, 2023*  
Donders, NL  
Collaborative Donders meeting between departments

**The interplay between grey and white matter structure in supporting cognitive development in typical development** *Oct, 2023*  
Donders, NL  
Department meeting

## Hackathons

**PhD et al., foster collaborations between PhDs all over the world** *June 2023*  
Padova, Italy  
Society for the Improvement of Psychological Science

## Workshops organized

**How to make your research greener ?** *2023*  
Nijmegen, NL  
organised for the Donders Discussion

**How to write a pre-registration ?** *2021*  
(virtual)  
organised for the Fresco association  
WEPASCO: serie of workshops for students and young researchers

## POSTERS

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**Organisation for Human Brain Mapping - OHBM** *June, 2026*  
Bordeaux, FR  
Histological measures provide information beyond MRI metrics to predict differences in IQ

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| <b>Flux Congress</b><br>The neural scars of adversity: how do different kinds of adversity predict the development of grey and white matter structure in two longitudinal neuroimaging cohorts? | <i>September, 2025</i><br>Dublin, IR |
| <b>European Society for Child and Adolescent Psychiatry</b><br>The neural scars of adversity: how do different kinds of adversity predict the development of grey and white matter structure?   | <i>June, 2025</i><br>Strasbourg, FR  |
| <b>European Student Conference on Behaviour and Cognition</b><br>Neurons' dendritic structure predicts differences in IQ but not through MRI cortical thickness                                 | <i>Nov, 2024</i><br>Paris, FR        |
| <b>CogSci</b><br>Does grey and white matter microstructure play complementary roles in supporting cognitive performance ?                                                                       | <i>Jul, 2024</i><br>Rotterdam, NL    |
| <b>Dutch Neuroscience Meeting</b><br>Interplay in the longitudinal development of grey and white matter structure in childhood and adolescence                                                  | <i>June, 2024</i><br>Tiel, NL        |
| <b>Donders/NIN meeting</b><br>Interplay in the longitudinal development of grey and white matter structure in childhood and adolescence                                                         | <i>October, 2023</i><br>Nijmegen, NL |
| <b>Organisation for Human Brain Mapping - OHBM</b><br>Interplay in the longitudinal development of grey and white matter structure in childhood and adolescence                                 | <i>July, 2023</i><br>Montreal, CA    |
| <b>Flux Society</b><br>Does grey and white matter microstructure play complementary roles in supporting cognitive performance ?                                                                 | <i>Sep, 2022</i><br>Paris, FR        |
| <b>Federation of European Neuroscience Societies - FENS</b><br>Does grey and white matter microstructure play complementary roles in supporting cognitive performance ?                         | <i>Jul, 2022</i><br>Paris, FR        |
| <b>L-CID conference "Development Matters"</b><br>Does grey and white matter microstructure play complementary roles in supporting cognitive performance ?                                       | <i>March, 2022</i><br>Leiden, NL     |
| <b>Donders Institute Poster session</b><br>Does grey and white matter microstructure play complementary roles in supporting cognitive performance ?                                             | <i>March, 2022</i><br>Nijmegen, NL   |
| <b>Colloquium of Young Researchers in Cognitive Sciences in France</b><br>Grey Matter Volume in Youths with Conduct Disorder: Effects of Callous-Unemotional Traits                             | <i>Mar, 2021</i><br>(virtual)        |

## PROFESSIONAL ACTIVITIES

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### Science Communication

**Committee Pint of Science Nijmegen** *2022 - 2025*  
International Science Festival

**Communications Committee Flux** *2022 - 2024*  
Flux: The Society for Developmental Cognitive Neuroscience

**Blog of Lifespan Cognitive Dynamic lab** *2021 - 2023*  
Creation and managing of the lab blog

### Initiatives

**Green Team Initiative CNS** *2023 - present*  
Working to make our department greener, e.g., measure of the carbon footprint related to travels for each group and award 10,000eur to the greener group

**PhD et al. - Founder** *2022 - 2024*  
Platform to foster collaborations between PhDs all over the world

### Professional Service

**SEA-SIG OHBM - Chair** *2023 - 2026*  
The sustainability and environmental actions group of OHBM foster sustainability for the neuroimaging community and the annual meeting

**Fresco - Honorary member** *2023 - present*  
Fresco - Federation of students and young researchers in cognitive science in France

**Halkes Women Faculty Network** *2023 - 2024*  
Foster equality and inclusion within the Radboud University

**Donders PhD Council** *2022 - 2023*  
Part of the DCMN council and the Mental Health Committee

**Fresco - Vice-president** *2020-2022*  
Fresco - Federation of students and young researchers in cognitive science in France  
Promoting and organising interaction and collaborative projects within the French ecosystem in cognitive science

**Cognivence - Treasurer** *2018-2019*  
Cognivence - Association of students and young researchers in cognitive sciences of Paris  
Organisation of the 18th edition of the « Forum des Sciences Cognitives »

## PUBLIC ENGAGEMENT

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**Talk during Pint of Science** *May, 2025*  
Layman talk on my PhD thesis for the festival  
Nijmegen, NL



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| <b>Scientific presentation in classroom</b><br>Presented my research to high school students (15-16 years old)    | <b>October, 2024</b><br>Nijmegen, NL        |
| <b>Scientific presentations in classroom</b><br>Presented the brain and my research to children (10-12 years old) | <b>April, 2024</b><br>Nijmegen, NL          |
| <b>Brain Olympiad Nijmegen</b><br>Organised an activity around 3D printed animals brains                          | <b>Feb, 2023 &amp; 2024</b><br>Nijmegen, NL |
| <b>Donders Open Day</b><br>Organised an activity around 3D printed animals brains                                 | <b>Oct, 2022</b><br>Nijmegen, NL            |
| <b>Cognitive Science Forum of Paris</b><br>Organised of the Forum to promote cognitive sciences                   | <b>Apr, 2022</b><br>Paris, FR               |
| <b>Cognitive Science Forum of Lille</b><br>Organised of the Forum to promote cognitive sciences                   | <b>Mar, 2022</b><br>Lille, FR               |
| <b>Cognitive Science Forum of Marseille</b><br>Organised of the Forum to promote cognitive sciences               | <b>Feb, 2022</b><br>Marseille, FR           |
| <b>Virtual escape game</b><br>Introduced concepts in cognitive science to a broad audience                        | <b>Apr, 2021</b>                            |
| <b>Cognitive Science Forum of Paris</b><br>Organised of the Forum to promote cognitive sciences                   | <b>Mar, 2019</b><br>Paris, FR               |

## TRAINING

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### Courses

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|-------------------------------------------------------------------------------|------------------|
| Scientific Outreach for Children. <i>Radboud University</i>                   | <i>Feb, 2024</i> |
| Analytic storytelling. <i>Radboud University</i>                              | <i>May, 2023</i> |
| Project Management for PhD candidate. <i>Radboud University</i>               | <i>May, 2023</i> |
| The Art of Presenting Science. <i>Radboud University</i>                      | <i>May, 2023</i> |
| Writing Scientific Articles. <i>Radboud University</i>                        | <i>Oct, 2022</i> |
| Education in a nutshell. <i>Radboud University</i>                            | <i>Mar, 2022</i> |
| Introduction to Longitudinal Structural Equation Modelling. <i>CenterStat</i> | <i>Dec, 2021</i> |
| Introduction to Structural Equation Modelling. <i>CenterStat</i>              | <i>Oct, 2021</i> |

### Workshops

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|-----------------------------------------------------------------------|------------------|
| Tractometry - OHBM course.                                            | <i>Jul, 2023</i> |
| Standing up for Science Workshop.                                     | <i>Dec, 2022</i> |
| Flux PreConference Workshop: Longitudinal Modelling.                  | <i>Sep, 2022</i> |
| Causal Inference from Longitudinal Data. <i>Jacobs/Cifar Workshop</i> | <i>Mar, 2022</i> |

## SKILLS

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| <b>Statistical procedures</b> | <b>Structural equation modelling, longitudinal structural equation modelling, mixed models, growth mixture models, generalized additive latent and mixed models</b> |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|

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|-----------------------------|------------------------------------------------------------------------------------------------------------------|
| <b>Statistical software</b> | <b>R, RStudio (expert), MPlus (beginner), SPSS (beginner)</b>                                                    |
| <b>Programming</b>          | <b>Python (intermediate), MATLAB (beginner), SPM12 (intermediate), FSL (beginner), FreeSurfer (intermediate)</b> |
| <b>Data collection</b>      | <b>Experience in collecting behavioral and neural data using electroencephalography (EEG)</b>                    |
| <b>Other skills</b>         | <b>Open Science, Clean and manage big datasets, Latex (basic) website development in R and wordpress</b>         |
| <b>Language</b>             | <b>French (native), English (C1), Dutch (B1)</b>                                                                 |

## REFERENCES

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Available upon request:

Prof Rogier Kievit, PhD supervisor (rogier.kievit@radboudumc.nl)

Dr Delia Fuhrmann, Collaborator and supervisor for my research visit (delia.fuhrmann@kcl.ac.uk)